Samuel Ruiperez-Campillo

sruiperez@ethz.ch | sruiperez@stanford.edu
(+34) 678 875 707
ORCID: 0000-0002-5425-4175
linkedin.com/in/samuel-ruiperez-campillo-8071a7177
./citations?user=xefOxB0AAAAJ&hl=en
Scopus ID: 57221782700
Nationality: Spanish **Birth:** 17 Feb 1998
Native, Bilingual, B2–C1, / Basic

Education

Ph.D. in Computer Science – ETH Zurich 2024 – Present

- Institute of Machine Learning – Medical Data Science Group
- Research Stays MIT Engineering and Harvard Medicine.
- Research Associate, Swiss AI Center; member, Swiss AI Initiative
- Topic: Generative AI, Explainable AI & Machine Learning in Medicine

Graduate Certificate in Applied Data Science – UC Berkeley 2022 – 2023

- GPA 4.0 / 4.0; *In Person* Coursework on Convex Optimization and ML for Healthcare
- Focus: Data Science & Machine Learning

M.Eng. in Bioinformatics & Computational Bioengineering – UC Berkeley 2022 – 2023

- GPA 4.0 / 4.0 (top 1 %); Master's thesis on Deep Learning in Bioinformatics
- Research project at Stanford University on AI & Signal Theory in Cardiology

M.Sc. in Biomedical Engineering – ETH Zurich 2020 – 2022

- GPA 5.5 / 6.0; Master's thesis on Variational Autoencoders in Cardiology (GPA 6.0 / 6.0)
- Internships at the Institute of ML and Internship at Stanford.
- Focus: Bioelectronics & Machine Learning

B.Sc. in Biomedical Engineering – Carlos III University 2016 – 2020

- GPA 9.2 / 10 (top 5%); Bachelor's thesis on Cardiac Multimodal Signal Processing
- Georgia Tech exchange (GPA 4.0 / 4.0); focus on Biophysics & Tissue Engineering

Research Positions

Research Fellow – Massachusetts Institute of Technology (Cambridge, MA, USA) Feb 2026 – Jul 2026

- Research fellow visitor at the Harvard-MIT Biomedical Engineering Center
- Work on Digital Health and AI for Cardiovascular Care

Research Associate – Swiss AI Center (Zurich, Switzerland) Feb 2024 – Present

- Member of the Swiss AI Initiative. Branch: Foundational Models for Healthcare
- Associated to my PhD Possition at the Institute of Machine Learning

Research Scientist Consultant – Stanford University (Stanford, CA, USA) Feb 2024 – Sep 2025

- Work on Machine Learning for electrophysiology at the Computational Arrhythmia Research Lab
- Falk Cardiovascular Research Center, Stanford Medicine.

Research Scientist – Stanford University (Stanford, CA, USA) Jun 2022 – Feb 2024

- Work on Machine Learning for electrophysiology at the Computational Arrhythmia Research Lab
- Falk Cardiovascular Research Center, Stanford Medicine.

Graduate Student Researcher (Capstone Project) – UC Berkeley (Berkeley, CA, USA) Jan 2022 – Jun 2022

- Work on deep learning in bioinformatics at the Life Language Processing Lab.

Graduate Student Researcher (Master Thesis) – ETH Zurich (Zurich, Switzerland) Aug 2021 – Dec 2021

- Work on Generative models for Health at the Institute of Machine Learning.

Graduate Student Researcher – ETH Zurich (Zurich, Switzerland) Jan 2021 – Jul 2021

- Work on neural technologies and signal processing at the Institute of Neuroinformatics

Research Intern – Polytechnic University of Valencia (Valencia, Spain) Sep 2019 – Aug 2020

- Part-time work on bio-signal processing at the ITACA Institute of Electrical Eng.

Research Fellow – UC3M (Madrid, Spain) Sep 2019 – Aug 2020

- Part-time work on biofluid mechanics at the Thermal & Fluids Engineering Department

Research Intern – Instituto Cajal, CSIC (Madrid, Spain) Jun 2019 – Sep 2019

- Work on microscopy and neuroengineering at Research Centre of the Spanish Government

Research Intern – Georgia Tech & Emory University (Atlanta, GA, USA) Aug 2018 – Jun 2019

- Cancer Immunotherapy & Microfluidics, Wallace H. Coulter Dept. of Biomedical Engineering

Industry Positions

Machine Learning Consultant – Physcade Inc. (Silicon Valley, CA, USA) Feb 2024 – Present

- Consulting services to the software team in a fast-growing, multi-million-dollar funded start-up
- Algorithm development, end-to-end ML pipeline design, AI safety & ethics, regulatory compliance

Machine Learning Engineer – Physcade Inc. (Silicon Valley, CA, USA) Jun 2022 – Feb 2024

- Integral member of the software team in a fast-growing, multi-million-dollar funded start-up
- Awarded company stock options as an early-stage employee
- Contributed to the pre-submission of an FDA application (granted Jun 2025), elevating the company's clinical impact
- Co-authored a patent for innovative cardiac catheter technology and built ML guidance software for clinicians

Co-founder – SWiiFT Inc. (Berkeley, CA, USA) Jan 2022 – Jun 2022

- Co-founded SWiiFT Inc., a dynamic start-up in Silicon Valley
- Selected for the elite 0.1% cohort of SkyDeck at UC Berkeley, highlighting high pre-seed potential

Teaching Experience

Teaching Assistant Comp. Sci. – Machine Learning for Healthcare (ETH Zurich, 6 ECTS) Jan 2026 – Jul 2026
Teaching Assistant Comp. Sci. – Seminar Topics in ML Research (ETH Zurich, 2 ECTS) Aug 2025 – Dec 2025
Teaching Assistant Comp. Sci. – Advanced Machine Learning (ETH Zurich, 10 ECTS) Aug 2025 – Dec 2025
Teaching Assistant Comp. Sci. – Machine Learning for Healthcare (ETH Zurich, 6 ECTS) Jan 2025 – Jul 2025
Teaching Assistant Comp. Sci. – Seminar Topics in ML Research (ETH Zurich, 2 ECTS) Aug 2024 – Dec 2024
Teaching Assistant Comp. Sci. – Advanced Machine Learning (ETH Zurich, 10 ECTS) Aug 2024 – Dec 2024
Teaching Assistant Comp. Sci. – Data Science for Medicine (ETH Zurich, 4 ECTS) Jan 2024 – Jul 2024

Journal Publications

[J1] S. Laguna*, A. Agostini*, A. Ryser, **S. Ruipérez-Campillo**, I. Cannistraci, M. Vandenhirtz, S. Mandt, N. Deperrois, F. Nooralahzadeh, M. Krauthammer, T. M. Sutter, and J. E. Vogt. "Structure is Supervision: Multiview Masked Autoencoders for Radiology." *Transactions on Machine Learning Research*, pp. 1–26, Apr 2026. doi: 10.48550/arXiv.2511.22294

[J2] M. Rodrigo, **S. Ruipérez-Campillo**, P. Ganesan, R. Feng, and S. M. Narayan. "The Impact of Catheter Configuration on The Mapping of Atrial Fibrillation." *Circulation: Arrhythmia and Electrophysiology*, e014061, Mar 2026. doi: 10.1161/CIRCEP.125.014061 (Editor in Chief's Featured Article of the Volume)

[J3] **S. Ruipérez-Campillo**, A. Ryser, T. M. Sutter, B. Deb, R. Feng, P. Ganesan, K. A. Brennan, A. J. Rogers, M. Z. H. Kolk, F. V. Y. Tjong, S. M. Narayan, and J. E. Vogt. "Reducing Diverse Sources of Noise in Ventricular Electrical Signals Using Variational Autoencoders." *Expert Systems with Applications*, 300:130185, Nov 2025. doi: 10.1016/j.eswa.2025.130185

- **[J4]** E. Ramírez, R. Alós, J. B. Tonko, **S. Ruipérez-Campillo**, M. A. Gsell, G. Plank, P. D. Lambiase, J. Millet, and F. Castells. "Ultra-high Resolution for Accurate Analysis in Cardiac Mapping: A Path Towards Fulfillment of Assumptions in Omnipolar Technology." *Computers in Biology and Medicine*, 201:111399, 2026. doi: 10.1016/j.compbiomed.2025.111399
- **[J5]** P. Pantelidis, **S. Ruipérez-Campillo**, J. E. Vogt, A. Antonopoulos, I. Gialamas, G. E. Zakynthinos, M. Spartalis, P. Dilaveris, J. Millet, T. G. Papaioannou, E. Oikonomou, and G. Siasos. "ECG-XPLAIM: eXPlainable Locally-adaptive Artificial Intelligence Model for Arrhythmia Detection from Large-Scale Electrocardiogram Data." *Frontiers in Cardiovascular Medicine*, 12, 2025. doi: 10.3389/fcvm.2025.1659971
- **[J6]** J. B. Tonko, G. Cabero-Vidal, **S. Ruipérez-Campillo**, E. Cabrera-Borrego, C. Lozano, E. Ramírez, J. Moreno, P. Sánchez-Millán, A. Chow, J. Jiménez-Jáimez, J. Millet, F. Castells, and P. D. Lambiase "Endo-Epicardial Electrical Disarray in Arrhythmogenic Cardiomyopathy with Ventricular Arrhythmias." *Heart Rhythm*, 2025. doi: 10.1016/j.hrthm.2025.09.012
- **[J7]** G. Putignano*, **S. Ruipérez-Campillo***, Z. Yuan, J. Millet, S. Guerrero-Aspizua. "Mathematical Models and Computational Approaches in CAR-T Therapeutics." *Frontiers in Immunology*, 16, 2025. doi: 10.3389/fimmu.2025.1581210
- **[J8]** J. B. Tonko, **S. Ruipérez-Campillo**, G. Cabero-Vidal, E. Cabrera-Borrego, C. Roney, J. Jiménez-Jáimez, J. Millet, F. Castells, and P. D. Lambiase. "Vector Field Heterogeneity as a Novel Omnipolar Mapping Metric for Functional Substrate Characterization in Scar-Related Ventricular Tachycardias." *Heart Rhythm*, May 2025. doi: 10.1016/j.hrthm.2024.10.066
- **[J9]** P. Ganesan, M. Pedron, R. Feng, A. J. Rogers, B. Deb, H. J. Chang, **S. Ruipérez-Campillo**, V. Srivastava, K. A. Brennan, W. R. Giles, T. Baykaner, P. Clopton, P. J. Wang, U. Schotten, D. E. Krummen, and S. M. Narayan. "Comparing Phenotypes for Acute and Long-Term Response to Atrial Fibrillation Ablation Using Machine Learning." *Circulation: Arrhythmia and Electrophysiology*, e012860, Mar 2025. doi: 10.1161/CIRCEP.124.012860 (Editor in Chief's Featured Article of the Volume)
- **[J10]** N. Deperrois, H. Matsuo, **S. Ruipérez-Campillo**, M. Vandenhirtz, S. Laguna, A. Ryser, K. Fujimoto, M. Nishio, T. M. Sutter, J. E. Vogt, J. Kluckert, T. Frauenfelder, C. Blüthgen, F. Nooralahzadeh, and M. Krauthammer. "RadVLM: A Multitask Conversational Vision-Language Model for Radiology." *arXiv* preprint arXiv:2502.03333, 2025. doi: 10.48550/arXiv.2502.03333 (*Under review in Nat. Digital Medicine*)
- **[J11]** M. Z. H. Kolk, **S. Ruipérez-Campillo**, A. A. M. Wilde, R. E. Knops, S. M. Narayan, and F. V. Y. Tjong. "Prediction of Sudden Cardiac Death Using Artificial Intelligence: Current Status and Future Directions." *Heart Rhythm*, Mar 2025. doi: 10.1016/j.hrthm.2024.09.003
- **[J12]** R. Feng*, K. A. Brennan*, Z. Azizi, J. Goyal, B. Deb, H. J. Chang, P. Ganesan, P. Clopton, M. Pedron, **S. Ruipérez-Campillo**, Y. B. Desai, H. De Larochellière, T. Baykaner, M. V. Perez, M. Rodrigo, A. J. Rogers, and S. M. Narayan. "Engineering of Generative Artificial Intelligence and Natural Language Processing Models to Accurately Identify Arrhythmia Recurrence." *Circulation: Arrhythmia and Electrophysiology*, 18(1):e013023, Jan 2025. doi: 10.1161/CIRCEP.124.013023
- **[J13]** P. Pantelidis, P. Dilaveris, **S. Ruipérez-Campillo**, A. Goliopoulou, A. Giannakodimos, P. Theofilis, R. De Lucia, O. Katsarou, K. Zisimos, K. Kalogeras, E. Oikonomou, and G. Siasos. "Hearts, Data, and Artificial Intelligence Wizardry: From Imitation to Innovation in Cardiovascular Care." *Biomedicines* 13, no. 5 (Apr 2025): 1019.
- **[J14]** **S. Ruipérez-Campillo**, M. Reiss, E. Ramírez, A. Cebrián, J. Millet, F. Castells. "Clustering and Machine Learning Framework for Medical Time Series Classification." *Biocybernetics and Biomedical Engineering*, 44(3), 521–533, 2024. doi: 10.1016/j.bbe.2024.07.005
- **[J15]** E. Ramírez, **S. Ruipérez-Campillo**, R. Casado-Arroyo, J. L. Merino, J. E. Vogt, F. Castells, and J. Millet. "The Art of Selecting the ECG Input in Neural Networks to Classify Heart Diseases: A Dual Focus on Maximizing Information and Reducing Redundancy." *Frontiers in Physiology*, Oct 2024. doi: 10.3389/fphys.2024.1452829
- **[J16]** P. Ganesan, R. Feng, B. Deb, F. V. Y. Tjong, A. J. Rogers, **S. Ruipérez-Campillo**, S. Somani, P. Clopton, T. Baykaner, M. Rodrigo, J. Zou, F. Haddad, M. Zaharia, and S. M. Narayan. "Novel Domain Knowledge-Encoding Algorithm Enables Label-Efficient Deep Learning for Cardiac CT Segmentation to Guide Atrial Fibrillation Treatment in a Pilot Dataset." *Diagnostics*, Jul 2024. doi: 10.3390/diagnostics14141538
- **[J17]** M. Z. H. Kolk, . Z. H. Kolk, **S. Ruipérez-Campillo**, C. P. Allaart, A. A. M. Wilde, R. E. Knops, S. M. Narayan, and F. V. Y. Tjong. "Multimodal Explainable Artificial Intelligence Identifies Patients with Non-Ischaemic Cardiomyopathy at Risk of Lethal Ventricular Arrhythmias." *Nature Scientific Reports*, Jun 2024. doi: 10.1038/s41598-024-65357-x
- **[J18]** E. Ramírez, **S. Ruipérez-Campillo**, F. Castells, R. Casado-Arroyo, J. Millet. "Novel Synchronization Method for Vectorcardiogram Reconstruction from ECG Printouts: A Comprehensive Validation Approach." *Biomedical Signal Processing and Control*, Sep 2024. doi: 10.1016/j.bspc.2024.106027
- **[J19]** L. Pancorbo, **S. Ruipérez-Campillo**, Á. Tormos, A. Guill, R. Cervigón, A. Alberola, F. J. Chorro, J. Millet, and F. Castells. "Vector Field Heterogeneity for Assessing Locally Disorganised Cardiac Wavefronts from High-Density Multielectrodes."

IEEE Open Journal of Engineering in Medicine and Biology, Jan 2024. doi: 10.1109/OJEMB.2023.3344349

- [**J20**] M. Z. H. Kolk, **S. Ruipérez-Campillo**, L. Alvarez-Florez, B. Deb, E. J. Bekkers, C. P. Allaart, A.-L. C. J. Van Der Lingen, P. Clopton, I. Išgum, A. A. M. Wilde, R. E. Knops, S. M. Narayan, and F. V. Y. Tjong. "Dynamic Prediction of Malignant Ventricular Arrhythmias Using Neural Networks in ICD Patients." *Lancet eBioMedicine*, 99, Jan 2024. doi: 10.1016/j.ebiom.2023.104937
- [**J21**] M. Z. H. Kolk, **S. Ruipérez-Campillo**, B. Deb, E. J. Bekkers, C. P. Allaart, A. J. Rogers, A.-L. C. J. Van Der Lingen, L. Alvarez Florez, I. Isgum, B. D. De Vos, P. Clopton, A. A. M. Wilde, R. E. Knops, S. M. Narayan, and F. V. Y. Tjong. "Optimizing Patient Selection for Primary Prevention Implantable Cardioverter-Defibrillator Implantation: Utilizing Multimodal Machine Learning to Assess Risk of Implantable Cardioverter-Defibrillator Non-Benefit." *Europace*, 25(9), Sep 2023. doi: 10.1093/europace/euad271
- [**J22**] M. Z. H. Kolk, B. Deb, **S. Ruipérez-Campillo**, N. K. Bhatia, P. Clopton, A. A. M. Wilde, S. M. Narayan, R. E. Knops, and F. V. Y. Tjong. "Machine Learning of Electrophysiological Signals for the Prediction of Ventricular Arrhythmias: Systematic Review and Examination of Heterogeneity Between Studies." *Lancet eBioMedicine*, Feb 2023. doi: 10.1016/j.ebiom.2023.104462
- [**J23**] **S. Ruipérez-Campillo**, M. Crespo, Á. Tormos, A. Guill, A. Cebrián, A. Alberola, J. Heimer, F. J. Chorro, J. Millet, and F. Castells. "Evaluation and Assessment of Clique Arrangements for the Estimation of Omnipolar Electrograms in High-Density Electrode Arrays: An Experimental Animal Model Study." *Physical and Engineering Sciences in Medicine*, 46, 1193–1204, Jun 2023. doi: 10.1007/s13246-023-01287-8
- [**J24**] P. Ganesan, B. Deb, R. Feng, M. Rodrigo, **S. Ruipérez-Campillo**, A. J. Rogers, P. Clopton, P. J. Wang, S. Zeemering, U. Schotten, W.-J. Rappel, and S. M. Narayan. "Quantifying a Spectrum of Clinical Response in Atrial Tachyarrhythmias Using Spatiotemporal Synchronization of Electrograms." *Europace*, May 2023. doi: 10.1093/europace/euad055
- [**J25**] R. Feng, B. Deb, P. Ganesan, F. Tjong, A. Rogers, **S. Ruipérez-Campillo**, S. Somani, M. Rodrigo, P. Clopton, J. Zou, M. Zaharia, S. Narayan. "Segmenting Computed Tomograms for Cardiac Ablation Using Machine Learning Leveraged by Domain Knowledge Encoding." *Frontiers in Cardiovascular Medicine*, Oct 2023. doi: 10.3389/fcvm.2023.1189293
- [**J26**] F. Castells, **S. Ruipérez-Campillo**, I. Segarra, R. Cervigón, R. Casado-Arroyo, J. L. Merino, and J. Millet. "Performance Assessment of Electrode Configurations for the Estimation of Omnipolar Electrograms from High-Density Arrays." *Computers in Biology and Medicine*, Mar 2023. doi: 10.1016/j.compbiomed.2023.106604
- [**J27**] **S. Ruipérez-Campillo**, S. Castrejón, M. Martínez, R. Cervigón, O. Meste, J. L. Merino, J. Millet, and F. Castells "Non-invasive Characterisation of Macroreentrant Atrial Tachycardia Types from a Vectorcardiographic Approach with the Slow Conduction Region as a Cornerstone." *Computer Programs and Methods in Biomedicine*, Mar 2022. doi: 10.1016/j.cmpb.2021.105932

* Indicates shared first authorship.

Patents

- [**P1**] R. C. Abad-Juan, S. Anbazhakan, **S. Ruipérez-Campillo**, M. Rodrigo-Bort, M. I. Alhusseini & S.M. Narayan, "System and Method for Mapping Activation Waves to Guide Treatment of Biological Rhythm Disorders." U.S. Patent 12,290,370, Physcade Inc., 2025. link: US12290370B1

Book Chapters

- [**B1**] K. O. Kristinsdóttir, **S. Ruipérez-Campillo** & Þ. Helgason. "Quantifying Spasticity: A Review." In *Stroke-Management Pearls*, 2023. doi: 10.5772/intechopen.112794

Models and Datasets

- [**M1**] Deperrois, H. Matsuo, **S. Ruipérez-Campillo**, M. Vandenhirtz, S. Laguna, A. Ryser, K. Fujimoto, M. Nishio, T. Sutter, J. Vogt, J. Kluckert, T. Frauenfelder, C. Bluethgen, F. Nooralahzadeh, M. Krauthammer. "RadVLM Instruction Dataset (version 1.0.0)." *PhysioNet*, 2025. RRID: SCR_007345. doi: 10.13026/et5g-h222
- [**M2**] Deperrois, H. Matsuo, **S. Ruipérez-Campillo**, M. Vandenhirtz, S. Laguna, A. Ryser, K. Fujimoto, M. Nishio, T. Sutter, J. Vogt, J. Kluckert, T. Frauenfelder, C. Bluethgen, F. Nooralahzadeh, M. Krauthammer. "RadVLM Model (version 1.0.0)." *PhysioNet*, 2025. RRID: SCR_007345. doi: 10.13026/50kn-p490

Machine Learning and Computer Vision Conference Papers

[MIDL 2026] B. Gundersen, N. Deperrois, **S. Ruipérez-Campillo**, T. M. Sutter, J. E. Vogt, M. Moor, F. Nooralahzadeh, and M. Krauthammer. "Enhancing Radiology Report Generation and Visual Grounding using Reinforcement Learning." *Medical Imaging with Deep Learning (MIDL)*, 2026 (online).

[MIDL 2026] J. Wiers, D. R. Wessels, L. Arts, **S. Ruipérez-Campillo**, M. Z. H. Kolk, F. V. Y. Tjong, and E. J. Bekkers. "Geometry-Aware Cardiac MRI Representation Learning with Equivariant Neural Fields." *Medical Imaging with Deep Learning (MIDL)*, 2026 (online). (Oral: Among top 8% of all submitted papers)

[MICCAI 2025] B. Chen, C. Vincent-Cuaz, L. A. Schoenpflug, M. Madeira, L. Fournier, V. Subramanian, S. Andani, **S. Ruiperez-Campillo**, J. E. Vogt, R. Luisier, D. Thanou, V. H. Koelzer, P. Frossard, G. Campanella & G. Rätsch. "Revisiting Automatic Data Curation for Vision Foundation Models in Digital Pathology." *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, Sep 2025. doi: 10.48550/arXiv.2503.18709 (Among top 9% of all submitted papers)

[ICML 2025] A. Agostini*, S. Laguna*, A. Ryser*, **S. Ruiperez-Campillo***, M. Vandenhirtz, N. Deperrois, F. Nooralahzadeh, M. Krauthammer, T. M. Sutter & J. E. Vogt. "Leveraging the Structure of Medical Data for Improved Representation Learning." *International Conference on Machine Learning* (Foundation Models for Health Workshop), Jul 2025.

[ICLR 2025] L. Erlarcher*, **S. Ruipérez-Campillo***, H. Michel, S. Wellmann, T. M. Sutter, E. Ozkan, J. E. Vogt. "Predicting Pulmonary Hypertension in Newborns: A Multi-View VAE Approach." *13th International Conference on Learning Representations* (AI for Children Workshop), Mar 2025 (online).

[ICLR 2024] **S. Ruipérez-Campillo**, A. Ryser, T. M. Sutter, R. Feng, P. Ganesan, B. Deb, K. A. Brennan, A. J. Rogers, M. Z. H. Kolk, F. V. Y. Tjong, S. M. Narayan, J. E. Vogt. "A Denoising VAE for Intracardiac Time Series in Ischemic Cardiomyopathy." *12th International Conference on Learning Representations* (Time-Series for Healthcare Workshop), Apr 2024 (online).

Awards, Fellowships and Distinctions

Editor in Chief's Featured Article of the Month (March) – in Circulation A&E	2026
Best Oral Presentation on Multimodal AI, Foundation Models – European Soc. of Cardiology (DCAI)	2025
Best Oral Presentation – Physionet Challenge 2025, Intl. Conf. Computing in Cardiology (**$300**)	2025
Paul Dudley White International Scholar Award – by the American Heart Association (AHA)	2025
Editor in Chief's Featured Article of the Month (February) – in Circulation A&E	2025
Clinical Translation Award – International Conference Computing in Cardiology (**$1 000**)	2024
Award to the Five Honorary Alumni – University Carlos III of Madrid	2024
Best Oral Presentation Award – Spanish National Society of General Medicine Congress (**€400**)	2023
Young Investigator Award (co-author) – e-Cardiology, ESC Congress 2023	2023
HRS Travel Award – Attendance grant for Heart Rhythm Society, San Francisco (**$1 500**)	2022
First Prize "Ferrán Mocholí" – Best Sci. Journal Article, ITACA Institute (UPV)	2022
Fung Excellence Scholarship – U.C. Berkeley (**$15 000**)	2021
First Prize José María Ferrero Corral – Spanish Society of Biomedical Eng. (**€600**)	2020
#1 Graduate in Bioengineering in Spain – Spanish Society of Academic Excellence	2020
"la Caixa" Excellence Fellowship – full funding for Master's at UC Berkeley(**$100 000**)†	2020
Rafael del Pino Excellence Fellowship – full funding for Masters at ETH Zurich (**€48 000**)†	2020
Engineering Faculty Distinction – One of the top 8 (top <1%) records at Carlos III University (**€1 000**)	2020
Best Poster Award – Spanish National Society of General Medicine Congress (**€500**)	2019
National Research Fellow Grant – Spanish Education Ministry (**€2 000**)	2019
Madrid Award for Academic Excellence – Region of Madrid (**€3 000**)	2019/18/17
Scholarship for Excellent Academic Performance for Intl. Exchange at Georgia Tech (**$34 000**)	2018
Honour Distinction and Grant – Pre-University Academic Performance (**€3 000**)	2017
Castilla-La Mancha Honours – Excellence of Records (**€3 000**)	2016

† Foundation fully covers tuition and living expenses for the corresponding master's programme.

Engineering & Computer Science Conference Papers

[C1] **S. Ruipérez-Campillo**, M. Rau, P. Ganesan, K. A. Brennan, R. Feng, S. Bandyopadhyay, A. J. Rogers, S. M. Narayan & J. E. Vogt. "Physics-Inspired Diffusion Probabilistic Models for Improved Denoising in Intracardiac Time Series." *IEEE EMBC*, Jul 2025, pp. 1–5. doi: 10.1109/EMBC58623.2025.11252692

[C2] L. Erlacher*, A. Agostini*, **S. Ruipérez-Campillo***, E. Ozkan, T. M. Sutter & J. E. Vogt. "SwissBeatsNet: A Multilead Masked Autoencoder for Chagas Disease Detection." *Computing in Cardiology*, 2025, p. 1, ISSN: 2325-887X. doi: 10.22489/CinC.2025.160.

[C3] E. Ramirez, J. Tonko, R. Alos, **S. Ruipérez-Campillo**, P. Lambiase, J. Millet & F. Castells. "Omnipolar Technology in Intracardiac signals: Are We Fulfilling the Assumptions?" *IEEE EMBC*, Jul 2025, pp. 1–5. doi: 10.1109/EMBC58623.2025.11252745

[C4] **S. Ruipérez-Campillo**, A. Ryser, T. M. Sutter, ..., S. M. Narayan. "Can Generative AI Learn Physiological Waveform Morphologies? A Study on Denoising Intracardiac Signals in Ischaemic Cardiomyopathy." *IEEE EMBC*, Jul 2024, pp. 1–4. doi: 10.1109/EMBC53108.2024.10782966

[C5] R. T. O. Quixal, **S. Ruipérez-Campillo**, F. Castells & J. Millet. "EEG Tensorization for CNN-Based Classification in Comatose Patients Following Cardiac Arrest." *IEEE EMBC*, Jul 2024, pp. 1–4. doi: 10.1109/EMBC53108.2024.10782869

[C6] **S. Ruipérez-Campillo**, G. Cabero, J. Tonko, J. Millet, P. Lambiase & F. Castells. "Identification of potential ablation targets for ventricular tachycardia using a novel omnipolar-based propagation organization metric." *IEEE Computing in Cardiology*, Sep 2024. doi: 10.22489/CinC.2024.261

[C7] R. Alós, E. Ramírez, **S. Ruipérez-Campillo**, ... & J. Millet. "Exploring ICA for differentiating intracavitary signals based on tissue composition." *IEEE Computing in Cardiology*, Sep 2024. doi: 10.22489/CinC.2024.379

[C8] E. Ramírez, **S. Ruipérez-Campillo**, R. Alós, F. Castells, R. Casado-Arroyo & J. Millet. "Quantifying ECG redundancy through mutual information analysis among leads and its application in CNNs." *IEEE Computing in Cardiology*, Sep 2024. doi: 10.22489/CinC.2024.324

[C9] L. Ronda Rute, E. Ramírez, J. Hejc, **S. Ruipérez-Campillo**, F. Castells, M. Pešl, Z. Stárek & J. Millet. "Characterization of cardiac tissue using high-density grid catheter: validation in porcine model." *IEEE Computing in Cardiology*, Sep 2024. doi: 10.22489/CinC.2024.358

[C10] R. Cervigón, M. L. Domínguez, S. Perea, M. Fernandez, **S. Ruipérez-Campillo**, F. Castells, J. Millet & Medtronic Ibérica. "Dynamic changes in episodes of atrial fibrillation as predictors of permanent atrial fibrillation using implantable device." *IEEE Computing in Cardiology*, Sep 2024. doi: 10.22489/CinC.2024.214

[C11] R. Cervigón, **S. Ruipérez-Campillo**, J. Millet & F. Castells. "Loneliness and Heart Rate in Older Adults." *MEDICON (IEEE)*, Sep 2023, pp. 195–203. doi: 10.1007/978-3-031-49062-0_22

[C12] R. T. O. Quixal, **S. Ruipérez-Campillo**, . . . , F. Castells & J. Millet. "Modelo Basado en Redes Neuronales Convolucionales y Altamente Conexas para la Ayuda al Diagnóstico de Pacientes en Coma tras Paro Cardíaco." *CASEIB*, Nov 2023. ISBN 978-84-17853-76-1, pp. 589–592

[C13] E. Ramírez Candela, R. T. O. Quixal, **S. Ruipérez-Campillo**, . . . , F. Castells & J. Millet. "3D CNN as an Approach to Predict Cerebral Performance of Comatose Patients." *IEEE Computing in Cardiology*, Sep 2023. doi: 10.22489/CinC.2022.424

[C14] R. T. O. Quixal, E. Ramírez-Candela, **S. Ruipérez-Campillo**, F. Castells & J. Millet. "Heterogeneity Quantification of Electrophysiological Signal Propagation in High-Density Multielectrode Recordings." *IEEE Computing in Cardiology*, Sep 2023. doi: 10.22489/CinC.2023.364

[C15] M. Pedrón, P. Ganesan, R. Feng, B. Deb, H. Chang, **S. Ruipérez-Campillo**, S. Somani, Y. Desai, A. J. Rogers, P. Clopton, S. M. Narayan. "Defining the Predictive Ceiling of Electrogram Features Alone for Predicting Outcomes from Atrial Fibrillation Ablation." *IEEE Computing in Cardiology*, Sep 2023. doi: 10.22489/CinC.2023.073

[C16] L. Pancorbo, **S. Ruipérez-Campillo**, F. Castells & J. Millet. "Heterogeneity Quantification of ECG Signal Propagation in HD Multielectrode Recordings." *IEEE Computing in Cardiology*, Sep 2023. doi: 10.22489/CinC.2023.261

[C17] E. Ramírez, **S. Ruipérez-Campillo**, F. Castells, R. Casado-Arroyo, J. Millet. "Synchronization of Conventional ECG Recordings for Accurate VCG Reconstruction." *IEEE Computing in Cardiology*, Sep 2023. doi: 10.22489/CinC.2023.218

[C18] M. Crespo-Aguirre*, **S. Ruipérez-Campillo***, . . . , J. Millet. "Assessment of the Inter-Electrode Distance Effect over the Omnipole with High Multielectrode Arrays." *IEEE EMBC*, Jul 2023. doi: 10.1109/EMBC40787.2023.10341063

[C19] **S. Ruipérez-Campillo**, F. Castells & J. Millet. "Robust Framework for Medical Time Series Classification and Application to Real Scenarios in Modern Bioengineering." *IEEE EMBC*, Jul 2023. doi: 10.1109/EMBC40787.2023.10340158

- **[C20]** I. Segarra, **S. Ruipérez-Campillo**, . . . , J. Millet. "Mini Peltier Cell Array System for the Generation of Controlled Local Epicardial Heterogeneities." *IEEE EMBC*, Jul 2023. doi: 10.1109/EMBC40787.2023.10340369
- **[C21]** **S. Ruipérez-Campillo**, F. Castells & J. Millet. "Metodología Robusta Basada en los Fundamentos del Machine Learning para la Clasificación de Señales Biomédicas." *CASEIB*, Nov 2022. ISBN 978-84-09-45972-8, pp. 31–34
- **[C22]** M. Crespo-Aguirre, **S. Ruipérez-Campillo**, . . . , J. Millet. "Estudio Comparativo con Señales Epicárdicas de las Lim. del Omnipolo con Multielectrodos de Alta Densidad." *CASEIB*, Nov 2022. ISBN 978-84-09-45972-8, pp. 138–141
- **[C23]** L. Pancorbo, **S. Ruipérez-Campillo**, . . . , J. Millet. "Cuantificación de la Heterogeneidad del Sustrato Electrofisiológico Cardiaco en Registros de Multielectrodos de Alta Densidad." *CASEIB*, Nov 2022. ISBN 978-84-09-45972-8, pp. 150–153
- **[C24]** I. Segarra, **S. Ruipérez-Campillo**, F. Castells & J. Millet. "Nuevo Método para el Análisis Independiente de la Orientación de Matrices Multielectrodo Equiespaciadas." *CASEIB*, Nov 2022. ISBN 978-84-09-45972-8, pp. 134–137
- **[C25]** M. L. Cardo, A. Chulián, **S. Ruipérez-Campillo**, J. Millet, F. Castells & R. Cervigón. "ECG Analysis to Study Social Connections in Older Cardiac Patients." *IEEE Computing in Cardiology*, Sep 2022. doi: 10.22489/CinC.2022.423
- **[C26]** R. Cervigón, E. Franco, **S. Ruipérez-Campillo**, C. Lozano, F. Castells & J. Moreno. "Autocorrelation Function for Predicting Arrhythmic Recurrences in Patients Undergoing Persistent Atrial Fibrillation Ablation." *IEEE Computing in Cardiology*, Sep 2022. doi: 10.22489/CinC.2022.424
- **[C27]** F. van Lieshout, R. C. Klein, M. Z. Kolk, K. van Geijtenbeek, R. Vos, **S. Ruipérez-Campillo**, S. Narayan, F. V. Y. Tjong. "Deep Learning for Ventricular Arrhythmia Prediction Using Fibrosis Segmentations on Cardiac MRI Data." *IEEE Computing in Cardiology*, Sep 2022. doi: 10.22489/CinC.2022.191
- **[C28]** I. Segarra, **S. Ruipérez-Campillo**, F. Castells & J. Millet. "Novel Method for Orientation-Independent Analysis in Equi-Spaced Multi-Electrode Arrays." *IEEE Computing in Cardiology*, Sep 2022. doi: 10.22489/CinC.2022.367
- **[C29]** R. C. Klein, F. van Lieshout, M. Z. Kolk, K. van Geijtenbeek, R. Vos, **S. Ruipérez-Campillo**, S. Narayan, F. V. Y. Tjong *et al.* "Weakly-Supervised Deep Learning for LV Fibrosis Segmentation in Cardiac MRI Using Image-Level Labels." *IEEE Computing in Cardiology*, Sep 2022. doi: 10.22489/CinC.2022.197
- **[C30]** **S. Ruipérez-Campillo**, J. Millet & F. Castells. "Classification of Atrial Tachycardia Types Using Dimensional Transforms of ECG Signals and Machine Learning." *IEEE Computing in Cardiology*, Sep 2022. doi: 10.22489/CinC.2022.349
- **[C31]** **S. Ruipérez-Campillo**, . . . , J. Millet & F. Castells. "Parametrización del Vectorcardiograma Auricular para la Caracterización de Distintos Tipos de Flutter." *CASEIB*, Nov 2020. ISBN 978-84-09-25491-0
- **[C32]** **S. Ruipérez-Campillo**, S. Castrejón, M. Cossiani, R. Cervigón, O. Meste, J. L. Merino, J. Millet & F. Castells. "Slow Conduction Regions as a Valuable Vectorcardiographic Parameter for the Non-Invasive Identification of Atrial Flutter Types." *IEEE Computing in Cardiology*, Sep 2020. doi: 10.22489/CinC.2020.442

Scientific Conference Talks

ESC-DCAI 2025 - *Enhancing stability in cardiac risk stratification with equivariant neural fields* — Berlin, DE, 2025

ESC-DCAI 2025 - *Contrastive learning to enrich ECG with cardiac MRI to predict structural features and cardiovascular disease* (Best Oral Presentation in Multimodal AI and Foundational Models Award) — Berlin, DE, 2025

ESC-DCAI 2025 - *Deep learning-enabled ECG fingerprinting of low-voltage substrate in atrial fibrillation patients* — Berlin, DE, 2025

AHA 2025 - *Artificial-Intelligence Based Smart Catheter Guidance for Atrial Fibrillation Ablation: LargeRegistry Validation* — New Orleans, US, 2025

AHA 2025 - *Artificial Intelligence-Based Optical Mapping Identifies Active Pulmonary Veins and Outcomesfrom Atrial Fibrillation Ablation* (Paul Dudley White International Award) — New Orleans, US, 2025

AHA 2025 - *Artificial Intelligence-Based Inference of Tissue Activation in Atrial Fibrillation from NoisyClinical Electrograms* — New Orleans, US, 2025

IEEE-CinC 2025 - *SwissBeatsNet: A Multilead Masked Autoencoder for Chagas Disease Detection* (Best Oral Presentation Award – Physionet Challenge 2025) — Sao Paulo, BR, 2025

ESC 2025 - *Large language models can estimate atrial fibrillation burden and infer progression without ecgs from the electronic healthcare record* — Madrid, ES, 2025

ESC 2025 - *Stratifying pulmonary hypertension severity in newborns from multi-view echocardiograms using variational autoencoders* — Madrid, ES, 2025

ESC 2025 - *Biophysics-inspired deep learning for improved denoising in ventricular signals in ischemic cardiomyopathy* Madrid, ES, 2025

EHRA 2025 - *Comparing artificial-intelligence based tracking of atrial fibrillation waves with clinical phenotypes in patients undergoing ablation* Vienna, AT, 2025

EHRA 2025 - *Artificial Intelligence Beat-Tracking Improves Regional Rate* Vienna, AT, 2025

EHRA 2025 - *Quantifying local cardiac substrate heterogeneity from high density recordings: an experimental study* Vienna, AT, 2025

ICLR 2025 (AI4CH) - *Predicting Pulmonary Hypertension in Newborns: A Multi-View VAE Approach* Singapore, SG, 2025

ICLR 2024 (TS4H) - *A Denoising VAE for Intracardiac Time Series in Ischemic Cardiomyopathy* Vienna, AT, 2024

IEEE-CinC 2024 - *Dentification of potential ablation targets for ventricular tachycardia using a novel omnipolar-based propagation organization metric* (Clinical Translation Award) Karlsruhe, DE, 2024

AHA 2024 - *Field of View of Atrial Electrodes by Endocardial and Epicardial Computer Mapping* Chicago, US, 2024

AHA 2024 - *Optimal Unipolar and Bipolar Electrode Separation for Recording AF in Patients* Chicago, US, 2024

ESC 2024 - *Quantifying Cardiac Tissue Heterogeneity Using HD Grid: Validation in Porcine Model* London, UK, 2024

EHRA 2024 - *Quantifying Local Cardiac Substrate Heterogeneity from High-Density Recordings* Berlin, DE, 2024

HRS 2024 - *Using Gen AI to Reduce Noise in EP Signals: Outperforming State-of-the-Art Filters* Boston, US, 2024

IEEE-EMBC 2024 - *Can Generative AI Learn Physiological Waveform Morphologies? A Study on Denoising Intracardiac Signals in Ischemic Cardiomyopathy* Orlando, US, 2024

IEEE-EMBC 2024 - *EEG Tensorization Enhances CNN-Based Outcome Classification in Comatose Patients Following Cardiac Arrest* Orlando, US, 2024

SEMG 2023 - *Reconstruction and Classification of Human Signals via Advanced Signal Processing & Artificial Intelligence* (Best Oral Presentation Award) Cuenca, ES, 2023

IEEE-EMBC 2023 - *Robust Framework for Medical Time-Series Classification and Application to Real Scenarios in Modern Bio-Engineering* Sydney, AU, 2023

IEEE-EMBC 2023 - *Assessment of the Inter-Electrode Distance Effect over the Omnipole with HD Arrays* Sydney, AU, 2023

ESC 2023 - *Improving Omnipolar Electrogram Reconstruction: An Animal Model Study* Amsterdam, NL, 2023

ESC 2023 - *Quantification of Local Het. from Act. Maps of Omnipolar Multielectrode Recordings* Amsterdam, NL, 2023

EHRA 2023 - *Study of the Omnipolar EGM Reconstruction for Robustness Against Wavefront Propagation in Epicardial Signals* Barcelona, ES, 2023

EHRA 2023 - *Novel Reconstruction Technique of Omnipolar Signals in High-Density Electrode Arrays* Barcelona, ES, 2023

AHA 2022 - *ML of Action-Potential Shape to Define Refractory Periods in Ischemic Cardiomyopathy* Chicago, US, 2022

AHA 2022 - *ECG-Derived Features Enable Prediction of Sudden Death in Ischemic Cardiomyopathy* Chicago, US, 2022

CASEIB 2022 - *Metodología Robusta de ML para la Clasificación de Señales Biomédicas: Tres Desafíos* Valladolid, ES, 2022

IEEE-CinC 2022 - *Reducing Noise in Atrial Fibrillation EGMs Using Autoencoder Neural Networks* Tampere, FI, 2022

IEEE-CinC 2022 - *Classification of Atrial Tachycardia Types via Dimensional Transforms & ML* Tampere, FI, 2022

ESC 2022 - *Reduction of Artifacts and Noise in Small Electrogram Datasets Using Transfer ML* Barcelona, ES, 2022

ESC 2022 - *Artificial Intelligence to Reduce Artifact in Cardiac Electrophysiological Signals* Barcelona, ES, 2022

IUPESM 2022 - *Agglomerative Nesting and Distance-Metric Analysis for Unsupervised Classification of Simulated Atrial Flutter Forms* Singapore, SG, 2022

HRS 2022 - *Reducing Diverse Types of Noise in Electrophysiological Signals Using ML Autoencoders* San Francisco, US, 2022

EHRA 2022 - *Defining Refractoriness in Single Atrial Beats Using Autoencoder Neural Networks* Copenhagen, DK, 2022

EHRA 2022 - *Noise Reduction in Electrophysiological Signals Using Transfer Machine Learning* Copenhagen, DK, 2022

BPS 2022 - *Dimensional Transform of Periodic Multichannel Signals to Enhance Unsupervised Classification of Atrial Flutter Loops* San Francisco, US, 2022

CASEIB 2020 - *Parametrización del VCG Auricular para la Caracterización de Distintos Tipos de AFl* Online, 2020

IEEE-CinC 2020 - *Slow Conduction Regions as a Valuable Vectorcardiographic Parameter for the Non-Invasive Identification of Atrial Flutter Types* Rimini, IT, 2020

SEMG 2019 - *A VCG Approach to Discriminate Between Different AFl Re-Entrant Circuits* Toledo, ES, 2019
(Best Poster Award)

Atrial Signals 2019 - *A VCG Approach to Discriminate Between Different AFl Re-Entrant Circuits* Bordeaux, FR, 2019

ICLR – International Conference on Learning Representations; AHA – American Heart Association Scientific Sessions; ESC – European Society of Cardiology; EHRA – European Heart Rhythm Association; HRS – Heart Rhythm Society; EMBC – IEEE Engineering in Medicine & Biology Conference; SEMG – Spanish Society of General Medicine National Congress; CASEIB – Congress of the Spanish Society of Biomedical Engineering; CinC – Computing in Cardiology; IUPESM – World Congress on Biomedical Engineering & Medical Physics; BPS – Biophysical Society Annual Meeting. DCAI – Digital Health and AI Summit

Other Conferences and Invited Positions

Invited Attendee – Machine Learning Summer School Arequipa, PE, 2023
Attendee to the MLSS 2025 (presented poster)

Excellence Fellow – Rafael del Pino Foundation, ELI Workshop Madrid, ES, 2024
Entrepreneurship, Leadership and Innovation (invited speaker)

Computational Genomics Summer Institute Los Angeles, US, 2024
Invited presentation at UCLA

Excellence Fellow – Rafael del Pino Foundation, ELI Workshop Madrid, ES, 2023
Entrepreneurship, Leadership and Innovation (invited speaker)

From Science to Society Conference in University of Cambridge Cambridge, UK, 2022
Honorary Guest and Session Chair

International Conference on Logistics in Industry Yekaterinb., RU, 2020
Invited delegate, BEST Europe

Madrid Engineering Week – IBM Case Study Madrid, ES, 2019
Invited participant, BEST Europe

International Conference on Computer Programming Istanbul, TR, 2019
Invited delegate, BEST Europe

International Conference on Tissue Engineering Bucharest, RO, 2018
Invited delegate, BEST Europe

UCLA: University of California, Los Angeles; BEST: Board of European Students of Technology

Student Supervision

Lucia Pancorbo, Harvard University – Master Thesis Harvard-ETH Fall 2025
Deep-Learned Frameworks to Enable Standard Neuroimaging Analyses from Single-Hemispheric Brain MRI

Nikolay Kormushev, ETH Zurich – Research Intern in Machine Learning Fall 2025
Representation Learning and Signal Processing for ECG P-Wave Analyses

Alexandre Reol, ETH Zurich and Uni. of Zurich – Master Thesis in Computer Science Fall 2025
Software and Outcome Prediction Model for Hypothermic Oxygenated Organ Perfusion

Giuseppe Cianci, University of Barcelona – Master Thesis in Artificial Intelligence Spring 2025
Multimodal ML for Risk-stratification of AF patients from ECGs

Julian Heidenreich, ETH Zurich – Master Semester Project in Mechanical Engineering Spring 2025
Deep Echocardiogram Models with Transfer Learning under Domain Shifts

Basile Morel, ETH Zurich – Master Semester Project in Computer Science Spring 2025
MAMBA Models of Cardiac Time Series

Nils Ebeling, ETH Zurich – Master Semester Project in Computer Science Fall 2024
Time- and Spectral Probabilistic Diffusion Models in Time Series Denoising

Lucas Erlacher, ETH Zurich – Master Thesis in Computer Science Fall 2024
Multimodal Variational Autoencoders for Diagnosing Pulmonary Hypertension in Newborns

Moritz Rau, ETH Zurich – Master Thesis in Statistics Spring 2024
Antithetic Sampling Enhanced Diffusion Models for Denoising Cardiac Signals

Community Outreach

Reviewer:

- *Journals:* Nature Communications, IEEE JBHI, IEEE TBME, Nature Digital Medicine, Computers in Biology and Medicine, Biomedical Signal Processing and Control.
- *Conferences:* NeurIPS, ICML, ICLR, IEEE-EMBC.

Conference/Workshop Organizer:

EurIPS 2025: Multimodal Representation Learning for Healthcare Workshop. Copenhagen DK, 2025
Dagstuhl Seminar: Generative AI in Cardiology Dagstuhl DE, 2026

Board and Institutional Service

e-Cardiology Working Group – European Society of Cardiology 2024 – 2026
Appointed Board-Nucleus Member (Switzerland delegate)
Section: Digital Health, AI and Engineering for Cardiology

Training in Other Fields

Music:

- Formal education in French Horn and performances with symphony orchestras in Spain, Europe and the USA.

Social and Voluntary Work:

- Organiser and supervisor of youth activities (Spain and UK).
- French-horn player in philanthropic concerts across Europe.
- Designed and built an arm-warmer prototype for cold-environment workers under Prof. Omer Inan (Georgia Tech).

Memberships

ESC E-Cardiology – Board Member (Zurich, CH)	2024 – Pres.
AHA – American Heart Association (Dallas, US)	2024 – 2025
ACC – American College of Cardiology (Washington DC, US)	2021 – Pres.
HRS – Heart Rhythm Society, Trainee Member (Washington DC, US)	2021 – Pres.
EHRA – European Heart Rhythm Association, Silver (Brussels, BE)	2021 – Pres.
ESC – European Society of Cardiology (Sophia Antipolis, FR)	2022 – Pres.
BPS – Biophysical Society, Graduate Member (Rockville, US)	2021 – Pres.
Cal BMES – Biomedical Engineering Society (Berkeley, US)	2021 – 2022
IEEE / IEEE EMBS – Student Member (Zurich, CH)	2020 – Pres.
SEIB – Spanish Society of Biomedical Engineering (Madrid, ES)	2020 – Pres.
ACE – Analytics Club at ETH Zurich (Zurich, CH)	2020 – Pres.
BMES – Georgia Tech Chapter (Atlanta, US)	2018 – 2019
BEST – Board of European Students of Technology, Conference Fellow (Valencia, ES)	2016 – 2019

International Research Projects

PERSPICATH – Project member, Spanish Ministry of Science and Innovation (EUR 121 000)	2023–2026
Q-Substrate – Project member, Spanish Ministry of Science and Innovation (EUR 73 000)	2020–2023

Skills

Python (NumPy, Pandas, Matplotlib, Seaborn, OpenCV, scikit-learn, PyTorch, Lightning, Weights & Biases, torchvision, Transformers, HuggingFace, Jupyter, TensorBoard), R (RStudio, tidyverse), MATLAB, Bash, SQL, Blender, LaTeX, git (GitHub, GitLab), Docker, High-Performance Computing (Slurm, MPI), Linux/Unix environments

Clinical Abstracts and Other Indexed Scientific Publications

[**A1**] **S. Ruipérez-Campillo**, G. Cinanci, D. Spreen, A. Luca, T. Kueffer, V. Schlageter, P. Badertscher, P. Krisai, N. Schaerli, J. Boeddinghaus, I. Strebel, M. Kuhne, C. Sticherling, J. E. Vogt, S. Knecht. "Deep Learning-enabled ECG Fingerprinting of Low-voltage Substrate in Atrial Fibrillation Patients." *European Heart Journal-Digital Health* 7(Suppl_1), 2026, ztaf143-123. doi: 10.1093/ehjdh/ztaf143.123

[**A2**] J. Wiers, D. Wessels, L. Alvarez-Florez, A. Bujalance Gomez, **S. Ruipérez-Campillo**, M. Kolk, E. Bekkers, F. Tjong. "Enhancing Stability in Cardiac Risk Stratification with Equivariant Neural Fields." *European Heart Journal-Digital Health* 7(Supplement_1), 2026, ztaf143-052. doi: 10.1093/ehjdh/ztaf143.052

[**A3**] A. Bujalance Gomez, L. A. Alvarez-Florez, F. R. Raijmakers, J. W. Wiers, **S. Ruipérez-Campillo**, M. Z. H. Kolk, E. J. B. Bekkers, F. V. Y. Tjong. "Contrastive Learning to Enrich ECG with Cardiac MRI to Predict Structural Features and Cardiovascular Disease." *European Heart Journal-Digital Health* 7(Supplement_1), 2026, ztaf143-053. doi: 10.1093/ehjdh/ztaf143.053

[**A4**] P. Pantelidis, **S. Ruipérez-Campillo**, V. C. Mystakidi, R. De Lucia, M. Spartalis, J. Millet, J. Vogt, P. Dilaveris, E. Oikonomou, G. Siasos. "Beyond Black-box Electrocardiogram Analysis: Saliency-guided Deep Learning for Differential Diagnosis of Tachyarrhythmias." *European Heart Journal-Digital Health* 7(Suppl_1), 2026, ztaf143-030. doi: 10.1093/ehjdh/ztaf143.030

[**A5**] **S. Ruipérez-Campillo**, M. Rau, P. Ganesan, R. Feng, K. A. Brennan, M. H. Kolk, F. V. Y. Tjong, A. J. Rogers, S. M. Narayan, J. E. Vogt. "Biophysics-inspired Deep Learning for Improved Denoising in Ventricular Signals in Ischemic Cardiomyopathy." *European Heart Journal* 46(Supplement_1), 2025. doi: 10.1093/eurheartj/ehaf784.4450

[**A6**] **S. Ruipérez-Campillo**, L. Erlarcher, H. Michel, S. Wellmann, T. M. Sutter, E. Ozkan, J. E. Vogt. "Stratifying Pulmonary Hypertension Severity in Newborns from Multi-view Echocardiograms Using Variational Autoencoders." *European Heart Journal* 46(Supplement_1), 2025. doi: 10.1093/eurheartj/ehaf784.4446

[**A7**] K. Brennan, R. Feng, J. Goyal, H. J. Chang, B. Deb, S. Bandyopadhyay, R. A. Ansari, V. Srivastava, P. Ganesan, **S. Ruipérez-Campillo**, P. Clopton, T. Baykaner, H. De Larochelliere, A. J. Rogers, S. M. Narayan. "Large Language Models Can Estimate Atrial Fibrillation Burden and Infer Progression Without ECGs from the Electronic Healthcare Record." *European Heart Journal* 46(Supplement_1), 2025. doi: 10.1093/eurheartj/ehaf784.4501

[**A8**] **S. Ruipérez-Campillo**, T. Fillon, S. M. Narayan. "Artificial Intelligence-Based Optical Mapping Identifies Active Pulmonary Veins and Outcomes from Atrial Fibrillation Ablation." *Circulation* 152(Suppl_3), 2025, A4363426–A4363426. doi: 10.1161/circ.152.suppl_3.4363426

[**A9**] S. Sadri, K. Brennan, S. Bandyopadhyay, P. Ganesan, Y. Desai, E. Peralta, R. Feng, C. Sillett, **S. Ruipérez-Campillo**, P. Wang, P. Clopton, A. J. Rogers, S. M. Narayan. "Longitudinal Evaluation of Anti-Arrhythmic Drug Use to Predict Hospitalization or Death in Patients with Ventricular Tachycardia." *Circulation* 152(Suppl_3), 2025, A4370694–A4370694. doi: 10.1161/circ.152.suppl_3.4370694

[**A10**] S. Bandyopadhyay, P. Ganesan, K. Brennan, **S. Ruipérez-Campillo**, R. A. Ansari, P. Clopton, A. Perino, P. Wang, E. Ashley, M. Perez, S. M. Narayan, A. J. Rogers. "Identifying Optimum ECG Features to Predict Sudden Cardiac Arrest at Varying Time Points Before the Event." *Circulation* 152(Suppl_3), 2025, A4369502–A4369502. doi: 10.1161/circ.152.suppl_3.4369502

[**A11**] **S. Ruipérez-Campillo**, T. Fillon, S. M. Narayan. "Artificial Intelligence-Based Inference of Tissue Activation in Atrial Fibrillation from Noisy Clinical Electrograms." *Circulation* 152(Suppl_3), 2025, A4367902–A4367902. doi: 10.1161/circ.152.suppl_3.4367902

[**A12**] S. Sadri, K. Brennan, S. Bandyopadhyay, Y. Desai, P. Ganesan, E. Peralta, R. Feng, C. Sillett, **S. Ruipérez-Campillo**, P. Wang, P. Clopton, A. J. Rogers, S. M. Narayan. "Large Language Models Detect Ventricular Tachycardia Recurrence in Clinical Notes and Enable Prediction of Clinical Outcomes at Scale." *Circulation* 152(Suppl_3), 2025, A4366827–A4366827. doi: 10.1161/circ.152.suppl_3.4366827

[**A13**] **S. Ruipérez-Campillo**, T. Fillon, C. Brodt, M. N. Faddis, S. M. Narayan. "Artificial-Intelligence Based Smart Catheter Guidance for Atrial Fibrillation Ablation: Large Registry Validation." *Circulation* 152(Suppl_3), 2025, A4357398–A4357398. doi: 10.1161/circ.152.suppl_3.4357398

[**A14**] S. Bandyopadhyay, S. Sadri, K. Brennan, P. Ganesan, P. Clopton, **S. Ruipérez-Campillo**, E. Peralta, C. Sillett, A. J. Rogers, S. M. Narayan. "AI-based Prediction of Mortality in Patients with Ventricular Tachycardia." *Circulation* 152(Suppl_3), 2025, A4370585–A4370585. doi: 10.1161/circ.152.suppl_3.4370585

[**A15**] P. Ganesan, E. Peralta, **S. Ruipérez-Campillo**, S. Bandyopadhyay, A. J. Rogers, H. J. Chang, K. Brennan, C. Sillett, P. Clopton, A. Perino, S. Niederer, S. M. Narayan. "Novel Foundation Models for Detecting and Generating Text Reports of Atrial Fibrillation from 12-lead ECGs in a Large Registry." *Circulation* 152(Suppl_3), 2025, A4370248–A4370248. doi: 10.1161/circ.152.suppl_3.4370248

[**A16**] R. Abad Juan, S. Anbazhakan, **S. Ruipérez-Campillo**, M. Rodrigo Bort, S. M. Narayan. "Artificial Intelligence Beat-Tracking Improves Regional Rate." *Europace* 27(Suppl 1), 2025, euaf085-076. doi: 10.1093/europace/euaf085.076

[**A17**] E. Ramírez, J. B. Tonko, R. Alos, **S. Ruipérez-Campillo**, J. Millet, P. Lambiase, F. Castells. "The Continuous Grid Model: A Novel Approach to Improve Spatial Resolution and Omnipolar Signal Reliability for Electroanatomical Mapping of Arrhythmogenic Substrates." *Europace* 27(Suppl 1), 2025, euaf085-075. doi: 10.1093/europace/euaf085.075

[**A18**] S. Anbazhakan, R. Abad Juan, **S. Ruipérez-Campillo**, M. Rodrigo Bort, S. M. Narayan. "Designing Artificial-Intelligence Based Tracking of Atrial Fibrillation Waves to Identify Ablation Response." *Europace* 27(Suppl 1), 2025, euaf085-319. doi: 10.1093/europace/euaf085.319

[**A19**] S. Anbazhakan, R. Abad Juan, **S. Ruipérez-Campillo**, M. Rodrigo Bort, S. M. Narayan. "Comparing Artificial-Intelligence Based Tracking of Atrial Fibrillation Waves with Clinical Phenotypes in Patients Undergoing Ablation." *Europace* 27(Suppl 1), 2025, euaf085-251. doi: 10.1093/europace/euaf085.251

[**A20**] P. Ganesan, R. Feng, M. Pedron, A. J. Rogers, B. Deb, H. J. Chang, **S. Ruipérez-Campillo**, V. Srivastava, K. Brennan, W. R. Giles, T. Baykaner. "Explainable Machine Learning Identifies that Need for Atrial Fibrillation Induction at Ablation Strongly Predicts Both Acute and Long-Term Outcomes from Ablation." *Heart Rhythm* 22(4), 2025, S739–S740. doi: 10.1016/j.hrthm.2025.03.1780

[**A21**] F. Raijmakers, **S. Ruipérez-Campillo**, E. Bekkers, L. Alvarez-Florez, R. Knops, S. M. Narayan, M. Kolk, F. V. Tjong. "Explainable Multimodal Feature Clustering with Artificial Intelligence for Risk Profiling in Malignant Ventricular Arrhythmias." *Heart Rhythm* 22(4), 2025, S540–S541. doi: 10.1016/j.hrthm.2025.03.1351

[**A22**] R. Abad, S. Anbazhakan, **S. Ruipérez-Campillo**, M. Rodrigo, M. N. Faddis, S. M. Narayan. "Improved Estimation of Rate in Atrial Fibrillation Using Artificial Intelligence Beat-Identification." *Heart Rhythm* 22(4), 2025, S577–S578. doi: 10.1016/j.hrthm.2025.03.1434

[**A23**] S. Anbazhakan, R. Abad, **S. Ruipérez-Campillo**, M. Rodrigo, M. N. Faddis, S. M. Narayan. "Artificial-Intelligence Based Tracking of Atrial Fibrillation Waves to Identify Sites of Acute Ablation Success." *Heart Rhythm* 22(4), 2025, S740. doi: 10.1016/j.hrthm.2025.03.1781

[**A24**] M. Rodrigo, **S. Ruipérez-Campillo**, S. M. Narayan. "Contrasting Unipolar, Bipolar and Omnipolar Catheters to Map Atrial Fibrillation in Patients." *Heart Rhythm* 22(4), 2025, S463–S464. doi: 10.1016/j.hrthm.2025.03.1188

[**A25**] R. Abad, S. Anbazhakan, **S. Ruipérez-Campillo**, M. Rodrigo, M. N. Faddis, S. M. Narayan. "Estimation of Rapid Rates in Atrial Fibrillation Using Artificial Intelligence and Dominant Frequency in Signals of Varying Quality." *Heart Rhythm* 22(4), 2025, S243. doi: 10.1016/j.hrthm.2025.03.510

[**A26**] S. Anbazhakan, R. Abad, **S. Ruipérez-Campillo**, M. Rodrigo, M. N. Faddis, S. M. Narayan. "PV-Dependent Versus Non-PV-Dependent Atrial Fibrillation Phenotypes Identified Using Artificial-Intelligence Based Wave Tracking." *Heart Rhythm* 22(4), 2025, S256–S257. doi: 10.1016/j.hrthm.2025.03.538

[**A27**] K. Brennan, R. Feng, J. Goyal, H. J. Chang, B. Deb, V. Srivastava, P. Ganesan, S. Bandyopadhyay, R. A. Ansari, **S. Ruipérez-Campillo**, P. Clopton. "Large Language Models Identify Atrial Fibrillation Progression on Unprecedented Scale." *Journal of the American College of Cardiology* 85(12_Supplement), 2025, 237. doi: 10.1016/S0735-1097(25)00722-3

[**A28**] R. Feng, K. Brennan, P. Ganesan, J. Goyal, B. Deb, Z. Azizi, H. J. Chang, **S. Ruipérez-Campillo**, P. Clopton, T. Baykaner, A. J. Rogers. "Prompt-Engineering Optimized ChatGPT as a General Tool for Analyzing Complex Electronic Health Records." *Journal of the American College of Cardiology* 85(12_Supplement), 2025, 120. doi: 10.1016/S0735-1097(25)00605-9

[**A29**] R. C. Abad Juan, S. Anbazhakan, **S. Ruipérez-Campillo**, M. Rodrigo, S. M. Narayan. "Estimating AF Rate in Patients Using AI Referenced to Tissue Action Potentials." *Circulation* 150(Suppl 1), 2024, A4136983. doi: 10.1161/circ.150.suppl_1.4136983

[**A30**] M. Rodrigo, **S. Ruipérez-Campillo**, S. M. Narayan. "Optimal Unipolar and Bipolar Electrode Separation for Recording AF in Patients." *Circulation* 150(Suppl 1), 2024, A4139988. doi: 10.1161/circ.150.suppl_1.4139988

[**A31**] M. Rodrigo, **S. Ruipérez-Campillo**, S. M. Narayan. "Field of View of Atrial Electrodes by Endocardial and Epicardial Computer Mapping." *Circulation* 150(Suppl 1), 2024, A4139975. doi: 10.1161/circ.150.suppl_1.4139975

[**A32**] R. C. Abad Juan, S. Anbazhakan, **S. Ruipérez-Campillo**, M. Rodrigo, S. M. Narayan. "AI Can Match AF Electrograms to Tissue Action Potentials in Patients." *Circulation* 150(Suppl 1), 2024, A4140503. doi: 10.1161/circ.150.suppl_1.4140503

[**A33**] S. Anbazhakan, R. C. Abad Juan, **S. Ruipérez-Campillo**, M. Rodrigo, S. M. Narayan. "AI-Based Tracking of AF Waves that Exit Pulmonary Veins Predicts Response to Ablation." *Circulation* 150(Suppl 1), 2024, A4138724. doi: 10.1161/circ.150.suppl_1.4138724

[**A34**] S. Anbazhakan, R. C. Abad Juan, **S. Ruipérez-Campillo**, M. Rodrigo, S. M. Narayan. "Atrial Fibrillation Wave Vectors Determined by AI Indicate Clinical Phenotypes." *Circulation* 150(Suppl 1), 2024, A4137246. doi: 10.1161/circ.150.suppl_1.4137246

[**A35**] E. Ramírez, **S. Ruipérez-Campillo**, F. Castells, R. Casado-Arroyo, J. Millet. "How Does Lead Selection Impact Diagnosis through AI? Implications for Continuous Monitoring Systems Using CNNs." *European Heart Journal* 45(Suppl 1), 2024, ehae666-3498. doi: 10.1093/eurheartj/ehae666.3498

[**A36**] J. B. Tonko, **S. Ruipérez-Campillo**, ... & P. Lambiase. "Vector Field Heterogeneity (VFH) as a Novel Quantitative Marker of Electrical Disarray in Arrhythmogenic Cardiomyopathy with VT." *European Heart Journal* 45(Suppl 1), 2024, ehae666-654. doi: 10.1093/eurheartj/ehae666.654

[**A37**] L. Ronda Rute, E. Ramírez, **S. Ruipérez-Campillo**, J. Millet, F. Castells, J. Hejc. "Quantifying Cardiac Tissue Heterogeneity Using High-Density Grid Catheter: Validation in a Porcine Model." *European Heart Journal* 45(Suppl 1), 2024, ehae666-337. doi: 10.1093/eurheartj/ehae666.337

[**A38**] M. Z. Kolk, **S. Ruipérez-Campillo**, R. Knops, S. M. Narayan, F. V. Tjong. "Multimodal Explainable AI Identifies Patients with Non-Ischaemic Cardiomyopathy at Risk of Lethal Ventricular Arrhythmias." *Heart Rhythm* 21(5), 2024, S15. doi: 10.1016/j.hrthm.2024.03.265

[**A39**] **S. Ruipérez-Campillo**, L. Pancorbo, E. Ramírez, F. J. Chorro, J. L. Merino, R. Casado-Arroyo, F. Castells, J. Millet. "Quantifying Local Cardiac Substrate Heterogeneity from High-Density Recordings: An Experimental Study." *Europace* 26(Suppl 1), 2024, euae102-650. doi: 10.1093/europace/euae102.650

[**A40**] **S. Ruipérez-Campillo**, T. M. Sutter, A. Ryser, R. Feng, P. Ganesan, K. Brennan, ... & S. M. Narayan. "Using Generative AI to Reduce Noise in Electrophysiological Signals: Outperforming State-of-the-Art Filters." *Heart Rhythm* 21(5), 2024, S212–S213. doi: 10.1016/j.hrthm.2024.03.688

[**A41**] M. Pedron, P. Ganesan, R. Feng, V. Srivastava, **S. Ruipérez-Campillo**, A. J. Rogers, B. Deb, H. J. Chang, K. Brennan, P. Clopton, S. M. Narayan. "Correlation of Electrical and Structural Predictors of Acute Ablation Outcome in Patients with Atrial Fibrillation." *Heart Rhythm* 21(5), 2024, S281. doi: 10.1016/j.hrthm.2024.03.787

[**A42**] B. Deb, P. Ganesan, H. De Larochellière, H. J. Chang, A. J. Rogers, R. Feng, **S. Ruipérez-Campillo**, T. Baykaner, N. K. Bhatia, S. M. Narayan. "Electrical-Structural Phenotypes of Three-Year Response to AF Ablation Revealed by Machine Learning." *Heart Rhythm* 21(5), 2024, S724. doi: 10.1016/j.hrthm.2024.03.1759

[**A43**] X. Ai, K. Brennan, R. Feng, J. Goyal, Z. Azizi, P. Clopton, B. Deb, P. Ganesan, H. J. Chang, **S. Ruipérez-Campillo** *et al.* "Detection of Recurrent VT in Unstructured Medical Notes Using NLP Augmented by Synthetic Notes." *Heart Rhythm* 21(5), 2024, S577–S578. doi: 10.1016/j.hrthm.2024.03.1454

[**A44**] K. Brennan, Z. Azizi, R. Feng, J. Goyal, X. Liu, P. Ganesan, **S. Ruipérez-Campillo** ... & S. M. Narayan. "Automated, Accurate Identification of Ventricular Tachycardia from Electronic Health Records Using NLP." *Journal of the American College of Cardiology* 83(13 Suppl), 2024, 2644. doi: 10.1016/S0735-1097(24)04634-5

[**A45**] P. Ganesan, M. Pedron, R. Feng, **S. Ruipérez-Campillo**, A. J. Rogers, B. Deb, ... & S. M. Narayan. "Separating Patients with Long-Term Success versus Acute Response from AF Ablation Using Explainable Machine Learning." *Circulation* 148(Suppl 1), 2023, A17861. doi: 10.1161/circ.148.suppl_1.17861

[**A46**] R. Feng, K. A. Brennan, Z. Azizi, J. Goyal, M. Pedron, H. J. Chang, P. Ganesan, **S. Ruipérez-Campillo** ... & S. M. Narayan. "Optimizing ChatGPT to Detect VT Recurrence from Complex Medical Notes." *Circulation* 148(Suppl 1), 2023, A16401. doi: 10.1161/circ.148.suppl_1.16401

[**A47**] **S. Ruipérez-Campillo**, F. Castells, R. Casado-Arroyo, J. L. Merino, J. Millet. "Improving Omnipolar Electrogram Reconstruction: An Animal Model Study." *European Heart Journal* 44(Suppl 2), 2023, ehad655-290. doi: 10.1093/eurheartj/ehad655.290

[**A48**] **S. Ruipérez-Campillo**, L. Pancorbo, F. Castells, M. Crespo, F. J. Chorro, J. L. Merino, R. Casado-Arroyo, J. Millet. "Quantification of Local Cardiac Substrate Heterogeneity from High-Density Recordings: An Experimental Study." *European Heart Journal* 44(Suppl 2), 2023, ehad655-328. doi: 10.1093/eurheartj/ehad655.328

[**A49**] R. Feng, P. Ganesan, B. Deb, H. J. Chang, **S. Ruipérez-Campillo**, S. Somani *et al.* "Predicting Success of Atrial Fibrillation Ablation: Comparing Machine Learning Approaches of Intracardiac Electrograms." *European Heart Journal* 44(Suppl 2), 2023, ehad655-517. doi: 10.1093/eurheartj/ehad655.517

[**A50**] **S. Ruipérez-Campillo**, F. Castells, M. Crespo *et al.* "Study of the Omnipolar EGM Reconstruction for Robustness Against Wavefront Propagation in Epicardial Signals." *Europace* 25(Suppl 1), 2023, euad122-662. doi: 10.1093/europace/euad122.662

[**A51**] **S. Ruipérez-Campillo**, F. Castells, I. Segarra, J. L. Merino, R. Casado-Arroyo, J. Millet. "Novel Reconstruction Technique of Omnipolar Signals in High-Density Electrode Arrays." *Europace* 25(Suppl 1), 2023, euad122-589. doi: 10.1093/europace/euad122.589

[**A52**] M. Z. Kolk, **S. Ruipérez-Campillo**, ... & F. V. Tjong. "Improved Patient Selection for Primary-Prevention ICD Implantation by Predicting ICD Non-Benefit Using Artificial Intelligence." *Europace* 25(Suppl 1), 2023, euad122-537. doi: 10.1093/europace/euad122.537

[**A53**] P. Ganesan, B. Deb, R. Feng, M. Rodrigo, **S. Ruipérez-Campillo**, ... & S. M. Narayan. "Spectrum of Clinical Response in Atrial Tachyarrhythmias Identified by Spatiotemporal Synchronization of Electrograms." *Circulation* 146(Suppl 1), 2022, A11304. doi: 10.1161/circ.146.suppl_1.11304

[**A54**] J. Goyal, B. Deb, T. Le Menestrel, H. J. Chang, F. V. Tjong, **S. Ruipérez-Campillo** ... & S. M. Narayan. "Multivariate Predictors of Long-Term Outcome from Ventricular Tachycardia Ablation in a Large Registry." *Circulation* 146(Suppl 1), 2022, A14115. doi: 10.1161/circ.146.suppl_1.14115

[**A55**] R. Feng, B. Deb, P. Ganesan, F. V. Tjong, A. J. Rogers, **S. Ruipérez-Campillo**, ... & S. M. Narayan. "Automatic Left Atrial Volume and Sphericity Index Calculation from Cardiac CT Using Computer Graphics Imaging and Deep Learning." *Circulation* 146(Suppl 1), 2022, A15312. doi: 10.1161/circ.146.suppl_1.15312

[**A56**] B. Deb, N. K. Bhatia, A. J. Rogers, T. Baykaner, H. J. Chang, **S. Ruipérez-Campillo**, ... & M. N. Viswanathan. "Machine Learning to Probe Variability in Patient Outcomes after AF Ablation – The Stanford AF Registry (START)." *Circulation* 146(Suppl 1), 2022, A12117. doi: 10.1161/circ.146.suppl_1.12117

[**A57**] **S. Ruipérez-Campillo**, B. Deb, A. Selvalingam, R. Feng, P. Ganesan, F. V. Tjong, ... & S. M. Narayan. "ECG-Derived Features Enable Prediction of Sudden Death in Ischemic Cardiomyopathy." *Circulation* 146(Suppl 1), 2022, A11648. doi: 10.1161/circ.146.suppl_1.11648

[**A58**] R. Feng, B. Deb, P. Ganesan, F. V. Tjong, **S. Ruipérez-Campillo**, ... & S. M. Narayan. "Improving Cardiac Segmentation for Atrial Fibrillation Ablation: A Prospective Trial of Machine-Learned Geometric Dissection vs Experts." *Circulation* 146(Suppl 1), 2022, A15172. doi: 10.1161/circ.146.suppl_1.15172

[**A59**] **S. Ruipérez-Campillo**, B. Deb, A. Selvalingam, R. Feng, ... & S. M. Narayan. "Machine Learning of Action Potential Shape to Define Refractory Periods in Ischemic Cardiomyopathy." *Circulation* 146(Suppl 1), 2022, A11651. doi: 10.1161/circ.146.suppl_1.11651

[**A60**] R. Feng, B. Deb, P. Ganesan, A. J. Rogers, **S. Ruipérez-Campillo**, ... & S. M. Narayan. "Automatic Left Atrial Segmentation from Cardiac CT Using Computer Graphics Imaging and Deep Learning." *European Heart Journal* 43(Suppl 2), 2022, ehac544-472. doi: 10.1093/eurheartj/ehac544.472

[**A61**] P. Ganesan, A. J. Rogers, B. Deb, R. Feng, **S. Ruipérez-Campillo**, F. V. Tjong, ... & S. M. Narayan. "Novel Electrogram Featurization Reveals a Spectrum of Response to Ablation from Atrial Tachycardia to Types of Atrial Fibrillation." *European Heart Journal* 43(Suppl 2), 2022, ehac544-471. doi: 10.1093/eurheartj/ehac544.471

[**A62**] **S. Ruipérez-Campillo**, B. Deb, R. Feng, ... & S. M. Narayan. "Artificial Intelligence to Reduce Artifact in Cardiac Electrophysiological Signals." *European Heart Journal* 43(Suppl 2), 2022, ehac544-422. doi: 10.1093/eurheartj/ehac544.422

[**A63**] **S. Ruipérez-Campillo**, B. Deb, R. Feng, ... & S. M. Narayan. "Reduction of Artifacts and Noise in Small Electrogram Datasets Without Manual Annotation Using Transfer Machine Learning." *European Heart Journal* 43(Suppl 2), 2022, ehac544-2976. doi: 10.1093/eurheartj/ehac544.2976

[**A64**] P. Ganesan, A. J. Rogers, B. Deb, R. Feng, M. Rodrigo, **S. Ruipérez-Campillo**, ... & S. M. Narayan. "Spatiotemporal Signatures of Response to Atrial Fibrillation Ablation." *European Heart Journal* 43(Suppl 2), 2022, ehac544-601. doi: 10.1093/eurheartj/ehac544.601

[**A65**] M. Z. Kolk, B. Deb, **S. Ruipérez-Campillo**, S. M. Narayan, F. V. Tjong *et al.* "Clinical Phenotyping of ICD Patients to Identify the Association of Remote Device Monitoring on Survival Benefit: A Cluster Analysis." *European Heart Journal* 43(Suppl 2), 2022, ehac544-738. doi: 10.1093/eurheartj/ehac544.738

[**A66**] M. Z. Kolk, B. Deb, **S. Ruipérez-Campillo**, P. Clopton, S. M. Narayan, R. Knops, F. V. Tjong. "Artificial Intelligence for the Prediction of Ventricular Arrhythmias and Sudden Cardiac Death: A Systematic Review and Meta-Analysis." *Cardiovascular Digital Health Journal* 3(4), 2022, S26. doi: 10.1016/j.cvdhj.2022.07.060

[**A67**] **S. Ruipérez-Campillo**, B. Deb, R. Feng, P. Ganesan, P. Clopton, A. J. Rogers, S. M. Narayan *et al.* "Defining Refractoriness in Single Atrial Beats Using Autoencoder Neural Networks." *EP Europace* 24(Suppl 1), 2022, euac053.614.

doi: 10.1093/europace/euac053.614

[**A68**] B. Deb, A. Selvalingam, M. Alhusseini, A. J. Rogers, P. Ganesan, R. Feng, P. Clopton, **S. Ruipérez-Campillo**, S. M. Narayan *et al.* "ML Physiological Signatures from the ECG Predict Sudden Death in Ischemic Cardiomyopathy." *EP Europace* 24(Suppl 1), 2022, euac053.021. doi: 10.1093/europace/euac053.021

[**A69**] **S. Ruipérez-Campillo**, B. Deb, R. Feng, P. Ganesan, P. Clopton, A. J. Rogers & S. M. Narayan. "Noise Reduction in Electrophysiological Signals Using Transfer Machine Learning." *EP Europace* 24(Suppl 1), 2022, euac053.125. doi: 10.1093/europace/euac053.125

[**A70**] B. Deb, A. Selvalingam, M. I. Alhusseini, A. J. Rogers, P. Ganesan, R. Feng, **S. Ruipérez-Campillo**, S. M. Narayan *et al.* "Machine-Learned Physiological Signatures of Sudden Cardiac Death." *Heart Rhythm* 19(5), 2022, S371–S372. doi: 10.1016/j.hrthm.2022.03.532

[**A71**] R. Feng, B. Deb, P. Ganesan, A. J. Rogers, **S. Ruipérez-Campillo**, S. M. Narayan *et al.* "Automatic Cardiac CT Segmentation Based on Machine Learning: Is Manual Annotation Required for Training?" *Heart Rhythm* 19(5), 2022, S365. doi: 10.1016/j.hrthm.2022.03.517

[**A72**] R. Feng, P. Ganesan, B. Deb, A. J. Rogers, **S. Ruipérez-Campillo**, S. M. Narayan *et al.* "Predicting Ablation Response in Persistent Atrial Fibrillation by Machine Learning of Global Atrial Electrogram Patterns." *Heart Rhythm* 19(5), 2022, S116. doi: 10.1016/j.hrthm.2022.03.809

[**A73**] **S. Ruipérez-Campillo**, B. Deb, R. Feng, S. M. Narayan *et al.* "Reducing Diverse Types of Noise in Electrophysiological Signals Using Machine-Learned Autoencoders." *Heart Rhythm* 19(5), 2022, S136. doi: 10.1016/j.hrthm.2022.03.845

[**A74**] R. Feng, B. Deb, P. Ganesan, S. Somani, A. J. Rogers, **S. Ruipérez-Campillo**, S. M. Narayan *et al.* "Applying Computer Graphics-Imaging to Identify Patient-Specific Cardiac Structures from Electroanatomic Shells with No Manual Input." *Heart Rhythm* 19(5), 2022, S84. doi: 10.1016/j.hrthm.2022.03.745

[**A75**] P. Ganesan, B. Deb, R. Feng, M. Rodrigo, **S. Ruipérez-Campillo**, S. M. Narayan *et al.* "Novel Detection of Atrial Tachycardia-Like Islands During Atrial Fibrillation Predicts Response to Ablation." *Heart Rhythm* 19(5), 2022, S17–S18. doi: 10.1016/j.hrthm.2022.03.122

[**A76**] B. Deb, P. Ganesan, R. Feng, N. K. Bhatia, A. J. Rogers, **S. Ruipérez-Campillo**, S. M. Narayan *et al.* "Unsupervised Machine Learning Identifies Phenotypes for Atrial Fibrillation that Predict Acute Ablation Success." *Journal of the American College of Cardiology* 79(9 Suppl), 2022, 51. doi: 10.1016/S0735-1097(22)01042-7

[**A77**] P. Ganesan, B. Deb, R. Feng, M. Rodrigo, **S. Ruipérez-Campillo**, S. M. Narayan *et al.* "Targeting Synchronized Electrogram Islands within Atrial Fibrillation for Ablation." *Journal of the American College of Cardiology* 79(9 Suppl), 2022, 3. doi: 10.1016/S0735-1097(22)00994-9

[**A78**] R. Feng, P. Ganesan, B. Deb, A. J. Rogers, **S. Ruipérez-Campillo**, S. M. Narayan *et al.* "A Morphological Operator-Based Approach to Automatically Separate and Label Left Atrium Body and Pulmonary Veins" *Journal of the American College of Cardiology* 79(9 Suppl), 2022, 1244. doi: 10.1016/S0735-1097(22)02235-5